

Haotong Luan

Shenzhen, China

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Summary

Third-year automation undergraduate working on underactuated robotic hands, environment-adaptive grasping, and mechanism design. Experience spans gripper development, visual-servoing control, and hardware prototyping, with interests in VLMs, VLAs, and WAMs for reliable robotic manipulation.

Education

Shenzhen Technology University – Future Technology School

B.ENG. IN AUTOMATION -- THIRD-YEAR UNDERGRADUATE

Shenzhen, China

Sep. 2024 – Present

- GPA: 84.17/100. Research focus: underactuated grippers, environment-adaptive robotic hands, and mechanism design.

Tsinghua University | Shenzhen X-Institute

P-SRT PROGRAM -- JOINT TRAINING IN UNDERACTUATED ROBOTICS

Shenzhen, China

Sep. 2025 – Present

- Advisor: Associate Prof. Wenzeng Zhang. Current stage: P-SRT; program pathway: X-idea → P-SRT → E-SRT → ORIC → SURF.

Research Experience

Shanghai AI Laboratory

RESEARCH ASSISTANT -- VISUAL-SERVOING CONTROL AND END-EFFECTOR DESIGN

Remote

Apr. 2026 – Present

- Working with Researcher Yuan Gao on visual-servoing control for robotic-arm grasping; responsible for the mechanical design of the arm-mounted gripper.

Shenzhen X-Institute, Tsinghua University

P-SRT RESEARCHER -- UNDERACTUATED ROBOTIC GRIPPERS

Shenzhen, China

Sep. 2025 – Present

- Independently proposed the dual-slide adaptive gripper concept and conducted mechanism analysis and feasibility studies under Associate Prof. Wenzeng Zhang.
- Designed linear-motion linkages, parallelogram mechanisms, and idle-stroke transmissions for reliable single-actuator grasping. Led DS Hand (first author, RAAI 2025); collaborated on idle-stroke and GLINT grippers.

Shenzhen Loop Area Institute

QUANTUM & AI WINTER SCHOOL

Shenzhen, China

Jan. 2026

- Studied MLP/LSTM/Transformer-based quantum variational circuit simulation with Prof. Peng Zhang (Tianjin University), including state fidelity optimization, mixed-state simulation, and physics-constrained training.

Publications

* Corresponding author. Submitted manuscripts are listed separately.

PEER-REVIEWED PUBLICATIONS AND ACCEPTED PAPERS

[P1] DPS Hand: A Dual-Parallelogram and Slot-Guided Gripper for Adaptive Grasping and Environmental Scooping

Haotong Luan, Haokai Ding, Tao Yang*, Wenzeng Zhang*

IEEE CASE 2026

Accepted; RAS major conference; Shenyang, China, Aug. 17–21, 2026.

[P2] PlaceRecover: Layout-Invariant Point Cloud Recovery for Robust 3D Perception under LiDAR Placement Shifts

Benwu Wang, Zihang Wang*, Wenkai Zhu, Haotong Luan, Dong Kong, Binghao Wang, Yiming Zhou, Kailin Lyu, Teng

IEEE/RSJ IROS 2026

Yan, Haoyu Wang, Yaohua Liu, Qiang Zhang

Accepted; top robotics conference; Pittsburgh, PA, USA, Sep. 27–Oct. 1, 2026.

[P3] LitPlan: A Replayable Workflow Contract for Scientific Literature Agents

Haotong Luan

Proceedings of the 1st Workshop on Interpreting Agent Behavior (IAB)

at the 40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026)

Sydney, Australia, Dec. 12, 2026.

[P4] CSD: Conformal Selective Diagnosis for Reliable Bearing Fault Identification Under Domain Shift

Xi Yu, Anran Lu, Keyi Chen, Haotong Luan*, Yuan Gao*

IEEE SMC 2026

Accepted; CCF C; Bellevue, WA, USA, Oct. 4–7, 2026. Corresponding author.

[P5] Domain-Semantic Subspace Stacking for Financial Fraud Detection

Haotong Luan, Minmin Chen, Zhenhui Lin, Haokai Ding, Yiwei Sheng*

ICIC 2026

Accepted; CCF C; Toronto, Canada, Jul. 22–26, 2026.

[P6] An Underactuated Dual-Slide Adaptive Robotic Hand with Strong Environmental Adaptability

Haotong Luan, Wenzeng Zhang*, Tao Yang*

RAAI 2025

Singapore, Dec. 18–20, 2025. DOI: 10.1109/RAAI67517.2025.11423404.

[P7] An Idle-Stroke Underactuated Gripper with Independent Double-Parallelogram Linkages for Pinching and Adaptive Grasping

Yizhe Chen, **Haotong Luan**, Haokai Ding, Wenzeng Zhang*
Singapore, Dec. 18–20, 2025. DOI: 10.1109/RAAI67517.2025.11423397. RAAI 2025

[P8] GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation

Haokai Ding, **Haotong Luan**, Tao Yang*, Wenzeng Zhang*
Paris, France, Nov. 24–26, 2025. DOI: 10.1109/ICCMA67641.2025.11369602. ICCMA 2025

JOURNAL MANUSCRIPTS SUBMITTED / UNDER REVIEW

[M1] Manuscript submitted to Advanced Intelligent Systems

Haotong Luan, Jie Zhang, Yuan Gao
Intelligent systems and robotics; JCR Q1. Submitted, 2026

[M2] Manuscript under review at Scientific Reports

Haotong Luan, Xi Yu, Anran Lu, Keyi Chen, Jianwu Chen
JCR Q1. Under review, 2026

Patents Pending

3 invention patents and 1 utility model patent filed and under review.

Honors & Awards

- 2026 **AIR Summer School Admission**, Institute for AI Industry Research, Tsinghua University.
- 2026 **CASE 2026 IEEE RAS Travel Support** (Travel Grant).
- 2026 **National-Level Project, Principal Investigator**, College Student Innovation and Entrepreneurship Training Program: “Zero-Lash Smart Grasping”.
- 2026 **Provincial-Level Project, Member**, College Student Innovation Training Program: “Multi-Core Fiber Dynamic Optical Trap for Single-Cell Pathology”.
- 2026 **SDG Global Summer School Selection** (tuition waived), Zhejiang University & Harbin Institute of Technology.
- 2026 **Quantum & AI Winter School Selection**, youngest of only 9 undergraduates among 70+ participants.
- 2024–2025 **Dean’s Special Award for Rising Talent, Zhenxiong**, Future Technology School, SZTU.
- 2025 **Crystal Prize for Scientific & Technological Innovation**, top 10 across SZTU.
- 2024, 2025 **Outstanding Student Cadre**, Shenzhen Technology University.
- 2025 **Excellence Prize**, “Challenge Cup” Student Entrepreneurship Competition, university round.
- 2025 **Excellence Prize**, “Challenge Cup” Extracurricular Academic Science and Technology Competition, university round.
- 2025 **Outstanding Volunteer**, 15th National Games of China and 12th National Games for Persons with Disabilities / 9th National Special Olympic Games.

Professional Service & Leadership

- 2026 **Conference Reviewer**, IEEE-RAS International Conference on Humanoid Robots (Humanoids), a top conference in humanoid robotics.
- 2026 **Conference Reviewer**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), a top robotics conference.
- 2026 **Workshop Reviewer**, IAB @ NeurIPS 2026 Workshop.
- 2024–Present **Head of Organization Department**, Future Technology School, SZTU (since Dec. 2024).
- 2024–2025 **Class Monitor & Youth League Secretary**, Sep. 2024–Oct. 2025.

Skills

Mechanism Design	Parallel-jaw coupled grippers, linear linkages, parallelogram linkages, underactuated and tendon-driven hands; SolidWorks (modeling, assembly, kinematic simulation); Bambu Lab 3D printing; prototyping and grasping experiments.
Programming	Python; basic deep learning for point clouds and quantum circuit simulation.
Academic Writing	IEEE journal/conference formatting and submission; AI-assisted literature review.
Languages	Chinese (native); English (CET-4: 580; IELTS scheduled Dec. 2026).